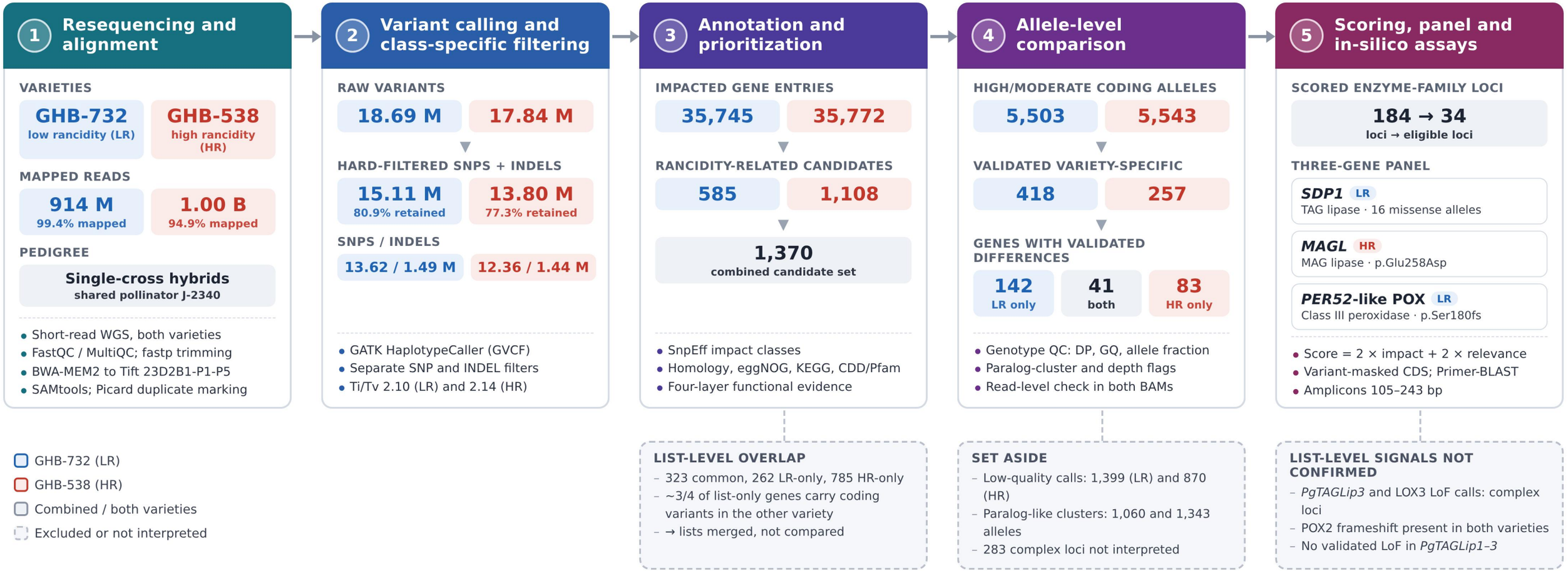

1
Resequencing and alignment
VARIETIES
GHB-732
low rancidity (LR)
GHB-538
high rancidity (HR)
MAPPED READS
914 M
99.4% mapped
1.00 B
94.9% mapped
PEDIGREE
Single-cross hybrids
shared pollinator J-2340
Short-read WGS, both varieties
FastQC / MultiQC; fastp trimming
BWA-MEM2 to Tift 23D2B1-P1-P5
SAMtools; Picard duplicate marking
2
Variant calling and class-specific filtering
RAW VARIANTS
18.69 M
17.84 M
HARD-FILTERED SNPS + INDELS
15.11 M
80.9% retained
13.80 M
77.3% retained
SNPS / INDELS
13.62 / 1.49 M
12.36 / 1.44 M
GATK HaplotypeCaller (GVCF)
Separate SNP and INDEL filters
Ti/Tv 2.10 (LR) and 2.14 (HR)
3
Annotation and prioritization
IMPACTED GENE ENTRIES
35,745
35,772
RANCIDITY-RELATED CANDIDATES
585
1,108
1,370
combined candidate set
SnpEff impact classes
Homology, eggNOG, KEGG, CDD/Pfam
Four-layer functional evidence
LIST-LEVEL OVERLAP
323 common, 262 LR-only, 785 HR-only
~3/4 of list-only genes carry coding variants in the other variety
→ lists merged, not compared
4
Allele-level comparison
HIGH/MODERATE CODING ALLELES
5,503
5,543
VALIDATED VARIETY-SPECIFIC
418
257
GENES WITH VALIDATED DIFFERENCES
142
LR only
41
both
83
HR only
Genotype QC: DP, GQ, allele fraction
Paralog-cluster and depth flags
Read-level check in both BAMs
SET ASIDE
Low-quality calls: 1,399 (LR) and 870 (HR)
Paralog-like clusters: 1,060 and 1,343 alleles
283 complex loci not interpreted
5
Scoring, panel and in-silico assays
SCORED ENZYME-FAMILY LOCI
184 → 34
loci → eligible loci
THREE-GENE PANEL
SDP1 LR
TAG lipase · 16 missense alleles
MAGL HR
MAG lipase · p.Glu258Asp
PER52-like POX LR
Class III peroxidase · p.Ser180fs
Score = 2 × impact + 2 × relevance
Variant-masked CDS; Primer-BLAST
Amplicons 105–243 bp
LIST-LEVEL SIGNALS NOT CONFIRMED
PgTAGLip3 and LOX3 LoF calls: complex loci
POX2 frameshift present in both varieties
No validated LoF in PgTAGLip1–3
GHB-732 (LR)
GHB-538 (HR)
Combined / both varieties
Excluded or not interpreted